THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry & Environmental Stewardship

Dear Dr. Way,

Please consider our paper, "Warmer and longer growing seasons increase growth for juvenile trees in a common garden" for publication as a research article in *Global Change Biology.*

What scientific question is addressed in this manuscript? Do trees grow more with longer and warmer seasons? To answer this question, we use a unique common garden data set of juvenile trees of four species sourced from multiple provenances. For the first time, we studied the effect of four growing season length metrics derived from leaf phenology on growth via tree rings.

What is/are the key finding(s) that answer this question? We demonstrate that the thermal season is a better predictor for growth than the calendar season. We also showed a larger response of growth to the—most often overlooked—end of season than to its start, further challenging the long-standing assumption that most growth happens in the spring.

Why is this work important and timely? Our work brings novel evidence on tree growth at a critical time where the role of the temperate forest carbon sink is paramount in this warming climate. We show that young deciduous trees may actually thrive under a warmer climate with critical implication on how we project and plan forest regeneration and range expansion.

Describe how your paper fits within the scope of GCB. The contribution of the temperate forest carbon sink relies on the assumption that longer and warmer growing seasons will maintain tree growth. Our work provides new and convincing evidence about this complex forest dynamic with important implications in carbon cycling, which we believe is of interest for Global Change Biology readers.

What biological AND global change aspects does it address? Our paper addresses tree growth for which two of the main drivers—temperature and water—are expected go through major major changes in their regime in the next decades. Our results provide new and striking evidence contributing to our understanding of how this critical biological process will respond to future climate.

What are the three most recently published papers that are relevant to this manuscript?
Dow *et al.* (2022), (Silvestro *et al.*, 2023), (Wolkovich *et al.*, 2025)

Provide the name and institutional email addresses of five potential reviewers.
Caroline Polgar—carolinepolgar@gmail.com
Amanda Gallinat—agallina@colby.edu
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All authors contributed to this work and approved this version for submission. The manuscript is 3847 words with a 283 word abstract, and 3 figures and is not under consideration elsewhere. We hope you find it suitable for publication in *Global Change Biology*, and look forward to hearing from you.

Sincerely,
Christophe Rouleau-Desrochers
MSc in Forestry
Department of Forest & Conservation Sciences
The University of British Columbia

Dow, C., Kim, A.Y., D'Orangeville, L., Gonzalez-Akre, E.B., Helcoski, R., Herrmann, V., Harley, G.L., Maxwell, J.T., McGregor, I.R., McShea, W.J., McMahon, S.M., Pederson, N., Tepley, A.J. & Anderson-Teixeira, K.J. (2022) Warm springs alter timing but not total growth of temperate deciduous trees. *Nature* **608**, 552–557.

Silvestro, R., Zeng, Q., Buttò, V., Sylvain, J.D., Drolet, G., Mencuccini, M., Thiffault, N., Yuan, S. & Rossi, S. (2023) A longer wood growing season does not lead to higher carbon sequestration. *Scientific Reports* **13**, 4059.

Wolkovich, E.M., Ettinger, A.K., Chin, A.R., Chamberlain, C.J., Baumgarten, F., Pradhan, K., Manzanedo, R.D. & Hille Ris Lambers, J. (2025) Why longer seasons with climate change may not increase tree growth. *Nature Climate Change* **15**, 1283–1292.